

12. The Riemann Solver of Osher

Osher's approximate Riemann solver is one of the earliest in the literature. The bases of the approach were communicated in the papers by Engquist and Osher in 1981 [120] and Osher and Solomon the following year [255]. Applications to the Euler equations were published later in a paper by Osher and Chakravarthy [253]. Since then the scheme has gained increasing popularity, particularly within the CFD community concerned with Steady Aerodynamics; see for example the works of Spekreijse [323], [324], Hemker and Spekreijse [167], Koren and Spekreijse [195], Qin et. al. [270], [271], [272], [273], [267], [268], [269]. One of the attractions of Osher's scheme is the smoothness of the numerical flux; the scheme has also been proved to be entropy satisfying and in practical computations it is seen to handle sonic flow well. A distinguishing feature of the Osher scheme is its performance near slowly-moving shock waves; see Roberts [280], Billett and Toro [40] and Arora and Roe [13]. The scheme is closely related to the Flux Vector Splitting approach described in Chap. 8 and, as Godunov's method of Chap. 6, it is a generalisation of the CIR scheme described in Chap. 5 for linear hyperbolic systems with constant coefficients. For a scalar conservation law, van Leer [395] studied in detail the relationship between the Osher scheme and some other Riemann solvers available at the time. Useful background material for reading this chapter is found in the previous Chaps. 2, 3, 5, 6, 8 and 9.

The derivation of the Osher intercell numerical flux depends on integration in phase space. Such operation involves the choice of *integration paths*, *intersection points* and *sonic points*. The integration paths are taken to be *integral curves* associated with the set of right eigenvectors and to date there are essentially two ways of *ordering* these integration paths. The most recent approach orders the integration paths such that these correspond to physically meaningful relations across wave families in *physical space*. In the current literature this is called *physical ordering* or *P-ordering*. Osher's original scheme utilises the ordering of paths that is precisely the inverse of the P-ordering; this is usually called *O-ordering*. Intersection and sonic points are computed via *Generalised Riemann Invariants*.

The purpose of this chapter is to present the Osher scheme in a way that can be directly applied to practical problems involving hyperbolic conservation laws. We first describe the principles behind Osher's method as

applied to any non-linear system of hyperbolic conservation laws. We then give detailed applications of the approach to a single scalar equation, to the isentropic equations of Gas Dynamics and to the split three-dimensional time dependent Euler equations for ideal gases. For one-dimensional applications, all the required information is found in this chapter and in Chap. 6.

12.1 Osher's Scheme for a General System

Here we give, in a self-contained manner, some of the basic aspects of the Osher scheme for a general non-linear system of hyperbolic conservation laws.

12.1.1 Mathematical Bases

Osher's approach to upwind differencing provides an approximation to the Godunov numerical flux of Chap. 6 and results from evaluating the physical flux $\mathbf{F}(\mathbf{U})$ at various states $\mathbf{U}_k$; these include the data states $\mathbf{U}_L$, $\mathbf{U}_R$, intersection points and sonic points. Consider a system of m hyperbolic conservation laws

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = \mathbf{0} \quad (12.1)$$

and the conservative scheme

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \frac{\Delta t}{\Delta x} \left[\mathbf{F}_{i-\frac{1}{2}} - \mathbf{F}_{i+\frac{1}{2}} \right] \quad (12.2)$$

to solve it numerically. The objective of this chapter is to provide an expression for the numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ following Osher's approach.

We assume (12.1) to be *strictly hyperbolic* with eigenvalues

$$\lambda_1(\mathbf{U}) < \lambda_2(\mathbf{U}) < \cdots < \lambda_m(\mathbf{U}) \quad (12.3)$$

and corresponding right eigenvectors

$$\mathbf{K}^{(1)}(\mathbf{U}), \mathbf{K}^{(2)}(\mathbf{U}), \dots, \mathbf{K}^{(m)}(\mathbf{U}).$$

From hyperbolicity, the Jacobian matrix

$$\mathbf{A}(\mathbf{U}) = \partial \mathbf{F} / \partial \mathbf{U} \quad (12.4)$$

is diagonalisable, that is

$$\mathbf{A}(\mathbf{U}) = \mathbf{K}(\mathbf{U}) \boldsymbol{\Lambda}(\mathbf{U}) \mathbf{K}^{-1}(\mathbf{U}), \quad (12.5)$$

where $\mathbf{K}(\mathbf{U})$ is the non-singular matrix whose columns are the right eigenvectors of $\mathbf{A}(\mathbf{U})$, that is

$$\mathbf{K}(\mathbf{U}) = \left[\mathbf{K}^{(1)}(\mathbf{U}); \mathbf{K}^{(2)}(\mathbf{U}); \dots; \mathbf{K}^{(m)}(\mathbf{U}) \right] \quad (12.6)$$

and $\mathbf{A}(\mathbf{U})$ is the diagonal matrix formed by the eigenvalues $\lambda_i(\mathbf{U})$

$$\mathbf{A}(\mathbf{U}) = \begin{pmatrix} \lambda_1(\mathbf{U}) & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & \lambda_m(\mathbf{U}) \end{pmatrix}. \quad (12.7)$$

See Sect. 2.3 of Chap. 2 and Sect. 3.2 of Chap. 3. As done in Sect. 2.3 of Chap. 2 for linear systems with constant coefficients, we introduce the following notation

$$\lambda_i^+(\mathbf{U}) = \max(\lambda_i(\mathbf{U}), 0); \quad \lambda_i^-(\mathbf{U}) = \min(\lambda_i(\mathbf{U}), 0) \quad (12.8)$$

to define diagonal matrices

$$\mathbf{A}^+(\mathbf{U}) = \begin{pmatrix} \lambda_1^+(\mathbf{U}) & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & \lambda_m^+(\mathbf{U}) \end{pmatrix}, \quad (12.9)$$

$$\mathbf{A}^-(\mathbf{U}) = \begin{pmatrix} \lambda_1^-(\mathbf{U}) & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & \lambda_m^-(\mathbf{U}) \end{pmatrix}, \quad (12.10)$$

and

$$|\mathbf{A}(\mathbf{U})| = \begin{pmatrix} |\lambda_1(\mathbf{U})| & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & |\lambda_m(\mathbf{U})| \end{pmatrix}. \quad (12.11)$$

We also introduce

$$|\mathbf{A}(\mathbf{U})| = \mathbf{K}(\mathbf{U})|\mathbf{A}(\mathbf{U})|\mathbf{K}^{-1}(\mathbf{U}). \quad (12.12)$$

But

$$|\lambda_i(\mathbf{U})| = \lambda_i^+(\mathbf{U}) - \lambda_i^-(\mathbf{U})$$

and hence

$$|\mathbf{A}(\mathbf{U})| = \mathbf{A}^+(\mathbf{U}) - \mathbf{A}^-(\mathbf{U}),$$

which if substituted in (12.12) gives

$$|\mathbf{A}(\mathbf{U})| = \mathbf{A}^+(\mathbf{U}) - \mathbf{A}^-(\mathbf{U}), \quad (12.13)$$

with

$$\mathbf{A}^+(\mathbf{U}) = \mathbf{K}(\mathbf{U})\mathbf{A}^+(\mathbf{U})\mathbf{K}^{-1}(\mathbf{U}) \quad (12.14)$$

and

$$\mathbf{A}^-(\mathbf{U}) = \mathbf{K}(\mathbf{U})\mathbf{A}^-(\mathbf{U})\mathbf{K}^{-1}(\mathbf{U}) . \quad (12.15)$$

These matrices produce a splitting of the Jacobian matrix as

$$\mathbf{A}(\mathbf{U}) = \mathbf{A}^+(\mathbf{U}) + \mathbf{A}^-(\mathbf{U}) , \quad (12.16)$$

where $\mathbf{A}^+(\mathbf{U})$ has positive or zero eigenvalues and $\mathbf{A}^-(\mathbf{U})$ has negative or zero eigenvalues. Note that this is a direct generalisation to non-linear systems of the Jacobian splitting for linear systems with constant coefficients performed in Sect. 5.4 of Chap. 5. See also Sect. 8.2.2 of Chap. 8 on the Flux Vector Splitting approach.

12.1.2 Osher's Numerical Flux

Osher's approach assumes that there exist vector-valued functions $\mathbf{F}^+(\mathbf{U})$ and $\mathbf{F}^-(\mathbf{U})$ that satisfy

$$\mathbf{F}(\mathbf{U}) = \mathbf{F}^+(\mathbf{U}) + \mathbf{F}^-(\mathbf{U}) \quad (12.17)$$

and

$$\frac{\partial \mathbf{F}^+}{\partial \mathbf{U}} = \mathbf{A}^+(\mathbf{U}) ; \quad \frac{\partial \mathbf{F}^-}{\partial \mathbf{U}} = \mathbf{A}^-(\mathbf{U}) . \quad (12.18)$$

If the initial data $\mathbf{U}_L$, $\mathbf{U}_R$ of the Riemann problem for the conservation laws (12.1) is denoted by

$$\mathbf{U}_0 \equiv \mathbf{U}_L = \mathbf{U}_i^n ; \quad \mathbf{U}_1 \equiv \mathbf{U}_R = \mathbf{U}_{i+1}^n , \quad (12.19)$$

then the corresponding numerical flux to be used in (12.2) is

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}^+(\mathbf{U}_0) + \mathbf{F}^-(\mathbf{U}_1) . \quad (12.20)$$

Sect. 8.2.2 of Chap. 8. Using the integral relations

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \mathbf{F}^-(\mathbf{U}_1) - \mathbf{F}^-(\mathbf{U}_0)$$

and

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^+(\mathbf{U}) d\mathbf{U} = \mathbf{F}^+(\mathbf{U}_1) - \mathbf{F}^+(\mathbf{U}_0) ,$$

we can express (12.20) in three different forms, namely

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}(\mathbf{U}_0) + \int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} , \quad (12.21)$$

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}(\mathbf{U}_1) - \int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^+(\mathbf{U}) d\mathbf{U} \quad (12.22)$$

and

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2} [\mathbf{F}(\mathbf{U}_0) + \mathbf{F}(\mathbf{U}_1)] - \frac{1}{2} \int_{\mathbf{U}_0}^{\mathbf{U}_1} |\mathbf{A}(\mathbf{U})| d\mathbf{U} . \quad (12.23)$$

Compare these flux formulae with those of Sect. 5.4.2 of Chap. 5 for linear systems with constant coefficients and with those for the Roe flux in Sect. 11.1.3 of Chap. 11.

The integration with respect to $\mathbf{U}$ in (12.21)–(12.23) is carried out in phase space $\mathbf{R}^m$. Elements of this vector space are vectors

$$\mathbf{U} = [u_1, u_2, \dots, u_m]^T , \quad (12.24)$$

whose components u_i are real numbers. In general, the integrals (12.21)–(12.23) *depend on the integration path chosen*. Osher's approach is to select particular integration paths *so as to make the actual integration tractable*.

12.1.3 Osher's Flux for the Single-Wave Case

The solution of the Riemann problem for (12.1) with data $\mathbf{U}_0, \mathbf{U}_1$ has m waves, in general. Osher's scheme utilises partial information on the solution to provide integration paths to evaluate the integrals (12.21)–(12.23), which in turn produce an expression for the numerical flux. Consider first the simplest case in which all waves in the solution of the Riemann problem, except for that associated with the eigenvalue $\lambda_j(\mathbf{U})$ and eigenvector $\mathbf{K}^{(j)}(\mathbf{U})$, are trivial. That is, the states $\mathbf{U}_0$ and $\mathbf{U}_1$ are connected by a single j -wave. Associated with any vector field $\mathbf{K}^{(k)}(\mathbf{U})$, there are *integral curves*. These have the property that their tangent lies in the direction of the eigenvector $\mathbf{K}^{(k)}(\mathbf{U})$, at any point $\mathbf{U}$ in phase space. For background on integral curves see [207] and [425]. An integration path $I_k(\mathbf{U})$ is now taken to be an integral curve of $\mathbf{K}^{(k)}(\mathbf{U})$ connecting $\mathbf{U}_0$ and $\mathbf{U}_1$. It is important to note here that the eigenvector $\mathbf{K}^{(k)}(\mathbf{U})$ and the eigenvalue $\lambda_k(\mathbf{U})$ associated with the relevant integration path *are not necessarily* those corresponding to the non-trivial j -wave family in question.

Suppose $I_k(\mathbf{U})$ is parameterised by $\mathbf{U}(\xi)$, $0 \leq \xi \leq \xi_1$, and

$$\mathbf{U}_0 = \mathbf{U}(0) , \quad \mathbf{U}_1 = \mathbf{U}(\xi_1) ; \quad (12.25)$$

then

$$\frac{d\mathbf{U}(\xi)}{d\xi} = \mathbf{K}^{(k)}(\mathbf{U}(\xi)) . \quad (12.26)$$

By performing a change of variables, utilising (12.25)–(12.26) and the fact that $\lambda_k^-(\mathbf{U})$ is an eigenvalue of $\mathbf{A}^-(\mathbf{U})$ with eigenvector $\mathbf{K}^{(k)}(\mathbf{U})$, we have

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \int_0^{\xi_1} \lambda_k^-[\mathbf{U}(\xi)] \mathbf{K}^{(k)}[\mathbf{U}(\xi)] d\xi . \quad (12.27)$$

We first consider the case in which the k -th field is *linearly degenerate*, that is $\lambda_k(\mathbf{U})$ is constant along $I_k(\mathbf{U})$; see Sect. 2.4.3 of Chap. 2 and Sect. 3.1.3

of Chap. 3. If $\lambda_k(\mathbf{U}) \geq 0 \quad \forall \mathbf{U}$ along $I_k(\mathbf{U})$, then $\lambda_k^-(\mathbf{U}) = 0$, see (12.8), and thus from (12.27) we have

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \mathbf{0} . \quad (12.28)$$

If $\lambda_k(\mathbf{U}) \leq 0 \quad \forall \mathbf{U}$ along $I_k(\mathbf{U})$, then manipulations of (12.27) give

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \int_0^{\xi_1} \frac{d\mathbf{F}}{d\mathbf{U}}[\mathbf{U}(\xi)] \frac{d\mathbf{U}}{d\xi} d\xi$$

and thus

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \mathbf{F}(\mathbf{U}_1) - \mathbf{F}(\mathbf{U}_0) . \quad (12.29)$$

Hence, if the k -field is linearly degenerate, use of (12.28) and (12.29) in the one-sided flux formula (12.21) gives the Osher's intercell flux as

$$\mathbf{F}_{i+\frac{1}{2}} = \begin{cases} \mathbf{F}(\mathbf{U}_0) & \text{if } \lambda_k \geq 0 \\ \mathbf{F}(\mathbf{U}_1) & \text{if } \lambda_k < 0 . \end{cases} \quad (12.30)$$

An entirely equivalent derivation of the Osher flux results from using the flux formulae (12.22) or (12.23). Note that in the case of a single wave associated with a linearly degenerate field, the Osher flux is identical to the Godunov flux if $j = k$.

We next consider the case in which the k -th field is *genuinely non-linear*, that is the eigenvalue $\lambda_k(\mathbf{U})$ is *monotone* along $I_k(\mathbf{U})$. This means that $\lambda_k(\mathbf{U})$ changes sign *at most once*, along $I_k(\mathbf{U})$. If $\lambda_k(\mathbf{U})$ does not change sign along $I_k(\mathbf{U})$ then this is simply like the linearly degenerate case above and the flux is given by (12.30). If $\lambda_k(\mathbf{U})$ changes sign at $\xi = \xi_S$, then there are two cases to consider. First assume

$$\lambda_k[\mathbf{U}(\xi)] \geq 0 , \quad \forall \xi \in [0, \xi_S] ; \quad \lambda_k[\mathbf{U}(\xi)] \leq 0 , \quad \forall \xi \in [\xi_S, \xi_1] .$$

Then (12.27) can be split into an integral between 0 and ξ_S and an integral between ξ_S and ξ_1 . The first integral is zero, as $\lambda_k^-(\mathbf{U}) = \min(\lambda_k(\mathbf{U}), 0)$ and $\lambda_k(\mathbf{U}) \geq 0$, see (12.8). The second integral gives

$$\int_{\xi_S}^{\xi_1} \frac{d\mathbf{F}}{d\mathbf{U}}[\mathbf{U}(\xi)] \frac{d\mathbf{U}}{d\xi} d\xi = \mathbf{F}(\mathbf{U}_1) - \mathbf{F}(\mathbf{U}_S) ,$$

where $\mathbf{U}_S = \mathbf{U}(\xi_S)$ and is called a *sonic point*. The case

$$\lambda_k[\mathbf{U}(\xi)] \leq 0 , \quad \forall \xi \in [0, \xi_S] ; \quad \lambda_k[\mathbf{U}(\xi)] \geq 0 , \quad \forall \xi \in [\xi_S, \xi_1]$$

can be treated in a similar way to give

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \mathbf{F}(\mathbf{U}_S) - \mathbf{F}(\mathbf{U}_0) .$$

Collecting results, for a single wave, we have

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \begin{cases} \mathbf{0} & \text{if } \lambda_k(\mathbf{U}) \geq 0 , \\ \mathbf{F}(\mathbf{U}_1) - \mathbf{F}(\mathbf{U}_0) & \text{if } \lambda_k(\mathbf{U}) \leq 0 , \\ \mathbf{F}(\mathbf{U}_1) - \mathbf{F}(\mathbf{U}_S) & \text{if } \lambda_k(\mathbf{U}_0) \geq 0 , \lambda_k(\mathbf{U}_1) \leq 0 , \\ & \lambda_k(\mathbf{U}_S) = 0 , \\ \mathbf{F}(\mathbf{U}_S) - \mathbf{F}(\mathbf{U}_0) & \text{if } \lambda_k(\mathbf{U}_0) \leq 0 , \lambda_k(\mathbf{U}_1) \geq 0 , \\ & \lambda_k(\mathbf{U}_S) = 0 . \end{cases}$$

By substituting these expressions into the one-sided flux formula (12.21) we obtain the Osher intercell flux, for the case in which the states $\mathbf{U}_0$ and $\mathbf{U}_1$ are connected by the single j -wave.

$$\mathbf{F}_{i+\frac{1}{2}} = \begin{cases} \mathbf{F}(\mathbf{U}_0) , & \lambda_k(\mathbf{U}) \geq 0 , \\ \mathbf{F}(\mathbf{U}_1) , & \lambda_k(\mathbf{U}) \leq 0 , \\ \mathbf{F}(\mathbf{U}_0) + \mathbf{F}(\mathbf{U}_1) - \mathbf{F}(\mathbf{U}_S) , & \lambda_k(\mathbf{U}_0) \geq 0 , \lambda_k(\mathbf{U}_1) \leq 0 , \\ & \lambda_k(\mathbf{U}_S) = 0 , \\ \mathbf{F}(\mathbf{U}_S) , & \lambda_k(\mathbf{U}_0) \leq 0 , \lambda_k(\mathbf{U}_1) \geq 0 , \\ & \lambda_k(\mathbf{U}_S) = 0 . \end{cases} \quad (12.31)$$

These results can be applied directly to any scalar, non-linear conservation law

$$u_t + f(u)_x = 0 . \quad (12.32)$$

12.1.4 Osher's Flux for the Inviscid Burgers Equation

Consider (12.32) with flux function

$$f(u) = \frac{1}{2} u^2 . \quad (12.33)$$

This gives the inviscid Burgers equation; see Sect. 2.4.2 of Chap. 2. Recall that the exact solution of the Riemann problem for (12.32)–(12.33) with data $u_L \equiv u_0$, $u_R \equiv u_1$ is

$$u(x/t) = \begin{cases} u_0 & \text{if } x/t \leq S = \frac{1}{2}(u_0 + u_1) \\ u_1 & \text{if } x/t > S \end{cases} \quad (12.34)$$

when $u_0 > u_1$ (shock case), and

$$u(x/t) = \begin{cases} u_0 & \text{if } x/t \leq u_0, \\ x/t & \text{if } u_0 \leq x/t \leq u_1, \\ u_1 & \text{if } x/t \geq u_1, \end{cases} \quad (12.35)$$

when $u_0 \leq u_1$ (rarefaction case). As, trivially, the only eigenvalue is the characteristic speed $\lambda = u$, direct application of (12.31) gives

$$\mathbf{F}_{i+\frac{1}{2}} = \begin{cases} f(u_0) & \text{if } u_0, u_1 > 0, \\ f(u_1) & \text{if } u_0, u_1 < 0, \\ f(u_0) + f(u_1) & \text{if } u_0 \geq 0 \geq u_1, \\ 0 & \text{if } u_0 \leq 0 \leq u_1. \end{cases} \quad (12.36)$$

At the sonic point $u_S = 0$ and thus $f(u_S) = 0$. Note that the Godunov flux $f_{i+\frac{1}{2}}^{god}$, obtained from the exact solution to the Riemann problem, is identical to Osher's flux (12.36), except in the case of a *transonic shock* ($u_0 \geq 0 \geq u_1$), where

$$f_{i+\frac{1}{2}}^{god} = \begin{cases} f(u_0) & \text{if } S \geq 0, \\ f(u_1) & \text{otherwise.} \end{cases}$$

For a full discussion on the relationship between the Godunov scheme, with the exact Riemann solver, and approximate Riemann solvers, including Osher's scheme, see the paper by van Leer [395].

12.1.5 Osher's Flux for the General Case

In the previous section we analysed Osher's numerical flux for the case of two states $\mathbf{U}_0, \mathbf{U}_1$ connected by a single wave, where an integration path was chosen to be tangential to a right eigenvector. For the general case of m wave families Osher chooses a set $\{I_k(\mathbf{U})\}$, $k = 1, \dots, m$, of *partial* integration paths such that $I_k(\mathbf{U})$ is tangential to $\mathbf{K}^{(k)}(\mathbf{U})$ and two successive paths $I_k(\mathbf{U})$ and $I_{k+1}(\mathbf{U})$ intersect at a *single point*

$$\mathbf{U}_{k/m} = I_k(\mathbf{U}) \cap I_{k+1}(\mathbf{U}) \quad (12.37)$$

in phase space, called an *intersection point*. The data points $\mathbf{U}_0$ and $\mathbf{U}_1$ are now to be interpreted as

$$\mathbf{U}_0 \equiv \mathbf{U}_{(k-1)/m}, \quad \mathbf{U}_1 \equiv \mathbf{U}_{k/m}.$$

The total integration path is the union of all partial paths, namely

$$I(\mathbf{U}) = \cup I_k(\mathbf{U}). \quad (12.38)$$

Fig. 12.1 illustrates a choice of integration paths for the case of a 3×3 hyperbolic system such as the one-dimensional, time dependent Euler equations.

The vectors $\mathbf{U}_0, \mathbf{U}_{1/3}, \mathbf{U}_{2/3}$ and $\mathbf{U}_1$ are vectors in phase space $\mathbf{R}^3$ and can be thought of as being the four constant states arising in the exact solution to the corresponding Riemann problem represented in physical space $x-t$ in Fig. 12.1. See Chap. 4 for details of the exact solution of the Riemann problem for the Euler equations. The points $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$ are the *intersection points* in phase space, and the points $\mathbf{U}_{S0}$ and $\mathbf{U}_{S1}$ are a representation of the potential *sonic points* that may arise from the non-linear fields associated with $\lambda_1(\mathbf{U})$ and $\lambda_3(\mathbf{U})$.

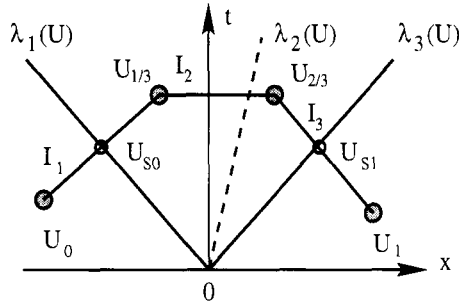


Fig. 12.1. Possible configuration of integration paths $I_k(\mathbf{U})$, intersection points $\mathbf{U}_{\frac{1}{3}}, \mathbf{U}_{\frac{2}{3}}$ and sonic points $\mathbf{U}_{S0}, \mathbf{U}_{S1}$ in physical space $x-t$ for a 3×3 system

For the general case, and assuming for the moment that the intersection points (12.37) and sonic points are known, the integration along $I(\mathbf{U})$ to evaluate (12.21), say, can now be performed by integrating along each partial integration path $I_k(\mathbf{U})$. But since these have been chosen to be tangential to the corresponding eigenvector $\mathbf{K}^{(k)}(\mathbf{U})$, the results of the previous section, see (12.31), can now be applied directly. The determination of the intersection points $\mathbf{U}_{k/m}$ and sonic points requires extra information about the solution of the Riemann problem. Traditionally, these have been determined by the use of Generalised Riemann Invariants, in at least two different ways, as we shall see.

12.2 Osher's Flux for the Isothermal Equations

The isothermal equations

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = \mathbf{0}, \quad (12.39)$$

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + a^2 \rho \end{bmatrix}, \quad (12.40)$$

have Jacobian matrix

$$\mathbf{A}(\mathbf{U}) = \begin{bmatrix} 0 & 1 \\ a^2 - u^2 & 2u \end{bmatrix} \quad (12.41)$$

with eigenvalues

$$\lambda_1(\mathbf{U}) = u - a, \quad \lambda_2(\mathbf{U}) = u + a, \quad (12.42)$$

and corresponding right eigenvectors

$$\mathbf{K}^{(1)}(\mathbf{U}) = \begin{bmatrix} 1 \\ u - a \end{bmatrix}, \quad \mathbf{K}^{(2)}(\mathbf{U}) = \begin{bmatrix} 1 \\ u + a \end{bmatrix}. \quad (12.43)$$

Recall that the sound speed a is constant here; for details on the isothermal equations see Sect. 1.6.2 of Chap. 1 and Sect. 2.4.1 of Chap. 2. The structure of the exact solution to the Riemann problem for (12.39)–(12.40) with initial data $\mathbf{U}_0, \mathbf{U}_1$ is depicted in Fig. 12.2 in the x - t plane, where the intersection point $\mathbf{U}_{\frac{1}{2}}$ is identified with the solution of the Riemann problem between the non-linear waves; potential sonic points are also shown.

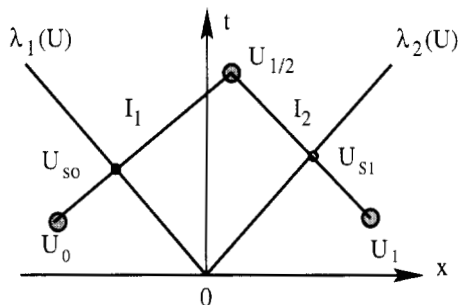


Fig. 12.2. Structure of the solution of the Riemann problem for the isothermal equations in physical space x - t . Integration paths are I_1 and I_2 , intersection point is $\mathbf{U}_{\frac{1}{2}}$, potential sonic points are $\mathbf{U}_{s0}$ and $\mathbf{U}_{s1}$; P-ordering

12.2.1 Osher's Flux with P-Ordering

There are two ways of ordering the integration paths in Osher's scheme, namely the *original* Osher ordering, or *O-ordering*, and the *physical* ordering following valid relations across waves in physical space, or *P-ordering*. First we apply *P-ordering*. It is easy to check, see Sect. 2.4.3 of Chap. 2, that the Riemann Invariants along the path $I_1(\mathbf{U})$, across the left wave in physical space, give

$$u_{\frac{1}{2}} + a \ln(\rho_{\frac{1}{2}}) = u_0 + a \ln(\rho_0). \quad (12.44)$$

Similarly, across the right wave

$$u_{\frac{1}{2}} - a \ln(\rho_{\frac{1}{2}}) = u_0 - a \ln(\rho_1). \quad (12.45)$$

The simultaneous solution for $\rho_{\frac{1}{2}}$ and $u_{\frac{1}{2}}$ is

$$u_{\frac{1}{2}} = \frac{1}{2}(u_0 + u_1) + \frac{1}{2}a \ln(\rho_0/\rho_1) , \quad (12.46)$$

$$\rho_{\frac{1}{2}} = \sqrt{\rho_0\rho_1} \exp \left[-\frac{u_1 - u_0}{2a} \right] \quad (12.47)$$

and thus the intersection point

$$\mathbf{U}_{\frac{1}{2}} = \left[\begin{array}{c} \rho_{\frac{1}{2}} \\ \rho_{\frac{1}{2}}u_{\frac{1}{2}} \end{array} \right] \quad (12.48)$$

is determined.

	Behaviour of $\lambda_1(\mathbf{U}) = u - a$	$\int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{2}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U}$
1	$u_0 - a \geq 0$, $u_{\frac{1}{2}} - a \geq 0$	$\mathbf{0}$
2	$u_0 - a \leq 0$, $u_{\frac{1}{2}} - a \leq 0$	$\mathbf{F}_{\frac{1}{2}} - \mathbf{F}_0$
3	$u_0 - a \geq 0$, $u_{\frac{1}{2}} - a \leq 0$	$\mathbf{F}_{\frac{1}{2}} - \mathbf{F}_{S0}$
4	$u_0 - a \leq 0$, $u_{\frac{1}{2}} - a \geq 0$	$\mathbf{F}_{S0} - \mathbf{F}_0$

Table 12.1. Evaluation of integral along integration path $I_1(\mathbf{U})$ with P-ordering, $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

The left and right sonic points $\mathbf{U}_{S0}$ and $\mathbf{U}_{S1}$ can easily be found by using the sonic conditions $u_{S0} = a$ from $\lambda_1 = u - a = 0$, $u_{S1} = -a$ from $\lambda_2 = u + a = 0$ and the Riemann Invariants. The result is

$$\rho_{S0} = \rho_0 \exp \left[\frac{u_0 - a}{a} \right] , \quad u_{S0} = a , \quad (12.49)$$

$$\rho_{S1} = \rho_1 \exp \left[-\frac{u_1 + a}{a} \right] , \quad u_{S1} = -a . \quad (12.50)$$

It is worth remarking at this stage, that the solution (12.46)–(12.47) for the intersection point is exact when the left and right waves are both rarefaction waves; in the general case it is an approximation. It is in fact the Two-Rarefaction approximation TRRS presented in Sect. 9.4.1 of Chap. 9 for the Euler equations.

	Behaviour of $\lambda_2(\mathbf{U}) = u + a$	$\int_{\mathbf{U}_{\frac{1}{2}}}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U}$
1	$u_{\frac{1}{2}} + a \geq 0, u_1 + a \geq 0$	$\mathbf{0}$
2	$u_{\frac{1}{2}} + a \leq 0, u_1 + a \leq 0$	$\mathbf{F}_1 - \mathbf{F}_{\frac{1}{2}}$
3	$u_{\frac{1}{2}} + a \geq 0, u_1 + a \leq 0$	$\mathbf{F}_1 - \mathbf{F}_{S1}$
4	$u_{\frac{1}{2}} + a \leq 0, u_1 + a \geq 0$	$\mathbf{F}_{S1} - \mathbf{F}_{\frac{1}{2}}$

Table 12.2. Evaluation of integral along integration path $I_2(\mathbf{U})$ with P-ordering, $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

Integration Along Partial Paths. In order to compute the Osher flux we use the one-sided flux formula

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}_0 + \int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U} , \quad (12.51)$$

where the integral is evaluated along each of the partial integration paths $I_1(\mathbf{U})$ and $I_2(\mathbf{U})$ shown in Fig. 12.2. For each case the integration is performed according to the local characteristic configuration. Tables 12.1 and 12.2 show the results for $I_1(\mathbf{U})$ and $I_2(\mathbf{U})$ respectively. As

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U} = \int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{2}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U} + \int_{\mathbf{U}_{\frac{1}{2}}}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U} , \quad (12.52)$$

strictly speaking, one should consider all 16 possible characteristic configurations (i, j) that result from Tables 12.1 and 12.2. Closer examination of all cases reveals that 4 possibilities are unrealisable. These are (1,2), (1,4), (4,2) and (4,4). For instance case (1,2) contains the requirements

$$u_{\frac{1}{2}} - a \geq 0 ; \quad u_{\frac{1}{2}} + a \leq 0 . \quad (12.53)$$

But $a > 0$ and thus these two conditions are contradictory. The remaining 12 cases can be tabulated as in the 3×4 Table 12.3, which gives the Osher intercell numerical flux $\mathbf{F}_{i+\frac{1}{2}}$. We separate conditions on the intersection point $\mathbf{U}_{\frac{1}{2}}$ (first column) from conditions on the data points $\mathbf{U}_0$ and $\mathbf{U}_1$ (top row). Note that in general $\mathbf{F}_{i+\frac{1}{2}}$ is a combination of physical flux values at

several points in phase space. In contrast, the Godunov flux obtained from the exact Riemann solver of Chap. 4 and from the approximate Riemann solvers of Chaps. 9, 10 and 11, consists of a *single value*. This has a bearing on the simplicity and computational efficiency of the schemes.

12.2.2 Osher's Flux with O-Ordering

In the original Osher scheme the ordering of the partial integration paths $I_k(\mathbf{U})$ is inverted. Fig. 12.3 illustrates the corresponding path configuration for the isothermal equations. The integration path connecting $\mathbf{U}_0$ to $\mathbf{U}_{\frac{1}{2}}$ is tangential to the eigenvector $\mathbf{K}^{(2)}(\mathbf{U})$ and that connecting $\mathbf{U}_{\frac{1}{2}}$ to $\mathbf{U}_1$ is tangential to $\mathbf{K}^{(1)}(\mathbf{U})$. Compare with Fig. 12.2. The O-ordering of the Osher scheme is used both for the integration paths as well as for the determination of the intersection point $\mathbf{U}_{\frac{1}{2}}$ and the sonic points $\mathbf{U}_{S0}, \mathbf{U}_{S1}$. The data state

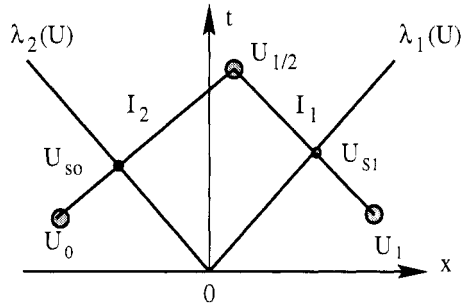


Fig. 12.3. Configuration of integration paths $I_k(\mathbf{U})$, intersection point $\mathbf{U}_{\frac{1}{2}}$ and potential sonic points $\mathbf{U}_{S0}, \mathbf{U}_{S1}$ in physical space $x-t$ when using the O-ordering

$\mathbf{U}_0$ is connected to $\mathbf{U}_{\frac{1}{2}}$ via the *right* (relation valid across the right wave) Riemann Invariant to give

$$u_{\frac{1}{2}} - a \ln(\rho_{\frac{1}{2}}) = u_0 - a \ln(\rho_0) .$$

Compare with (12.44). The data point $\mathbf{U}_1$ is connected to $\mathbf{U}_{\frac{1}{2}}$ via the *left* (relation valid across the left wave) Riemann Invariant to produce

$$u_{\frac{1}{2}} + a \ln(\rho_{\frac{1}{2}}) = u_1 + a \ln(\rho_1) .$$

Compare with (12.45). The resulting solution is

$$u_{\frac{1}{2}} = \frac{1}{2}(u_0 + u_1) + \frac{1}{2}a \ln(\rho_1/\rho_0) , \quad (12.54)$$

$$\rho_{\frac{1}{2}} = \sqrt{\rho_0 \rho_1} \exp \left[\frac{u_1 - u_0}{2a} \right] \quad (12.55)$$

	$u_0 - a \geq 0$ $u_1 + a \geq 0$	$u_0 - a \geq 0$ $u_1 + a \leq 0$	$u_0 - a \leq 0$ $u_1 + a \leq 0$	$u_0 - a \leq 0$ $u_1 + a \geq 0$
$u_{\frac{1}{2}} - a \geq 0$	$\mathbf{F}_0$	$\mathbf{F}_0 + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{S0} + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{S0}$
$u_{\frac{1}{2}} - a \leq 0$ $u_{\frac{1}{2}} + a \geq 0$	$\mathbf{F}_0 + \mathbf{F}_{\frac{1}{2}}$ $-\mathbf{F}_{S0}$	$\mathbf{F}_0 + \mathbf{F}_{\frac{1}{2}}$ $-\mathbf{F}_{S0} + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{\frac{1}{2}} + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{\frac{1}{2}}$
$u_{\frac{1}{2}} + a \leq 0$	$\mathbf{F}_0 + \mathbf{F}_{S1}$ $-\mathbf{F}_{S0}$	$\mathbf{F}_0 + \mathbf{F}_1$ $-\mathbf{F}_{S0}$	$\mathbf{F}_1$	$\mathbf{F}_{S1}$

Table 12.3. Osher's intercell flux for the isothermal equations using P-ordering, $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

and thus the intersection point $\mathbf{U}_{\frac{1}{2}}$ in (12.48) is determined. The left and right sonic points are evaluated using the sonic conditions $\lambda_2 = u + a = 0$ and $\lambda_1 = u - a = 0$ and the right and left Riemann Invariants, respectively. The result is

$$u_{S0} = -a, \quad \rho_{S0} = \rho_0 \exp \left[-\frac{u_0 + a}{a} \right], \quad (12.56)$$

$$u_{S1} = a, \quad \rho_{S1} = \rho_1 \exp \left[\frac{u_1 - a}{a} \right]. \quad (12.57)$$

Compare solutions (12.54)–(12.57) with (12.46)–(12.50) obtained with the P-ordering of the Osher scheme. Note that the solution for the intersection point given by the O-ordering has no physical meaning as a solution for the *Star Region* between the left and right wave families. If the exact solution of the Riemann problem consists of two rarefactions, then solution (12.47) is exact, in which case the density is expected to decrease; solution (12.55) gives an increase in density in the *Star Region*, which is obviously incorrect. We expand on this point when dealing with the Euler equations later in Sect. 12.3.3.

Integration along the path $I_2(\mathbf{U})$ connecting $\mathbf{U}_0$ to $\mathbf{U}_{\frac{1}{2}}$ produces the results of Table 12.4. The results of integrating along the path $I_1(\mathbf{U})$ connecting $\mathbf{U}_{\frac{1}{2}}$ to $\mathbf{U}_1$ are given in Table 12.5.

Out of the 16 possible combinations 12 cases are realisable. Table 12.6 gives the Osher numerical flux $\mathbf{F}_{i+\frac{1}{2}}$ for all 12 cases. Compare with Table 12.3 in which the flux is computed using the P-ordering.

	Behaviour of $\lambda_2(\mathbf{U}) = u + a$	$\int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{2}}} \mathbf{A}^-(\mathbf{U}) d\mathbf{U}$
1	$u_0 + a \geq 0, u_{\frac{1}{2}} + a \geq 0$	$\mathbf{0}$
2	$u_0 + a \leq 0, u_{\frac{1}{2}} + a \leq 0$	$\mathbf{F}_{\frac{1}{2}} - \mathbf{F}_0$
3	$u_0 + a \geq 0, u_{\frac{1}{2}} + a \leq 0$	$\mathbf{F}_{\frac{1}{2}} - \mathbf{F}_{S0}$
4	$u_0 + a \leq 0, u_{\frac{1}{2}} + a \geq 0$	$\mathbf{F}_{S0} - \mathbf{F}_0$

Table 12.4. Integration along path $I_2(\mathbf{U})$ connecting $\mathbf{U}_0$ to $\mathbf{U}_{\frac{1}{2}}$ (O-ordering), $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

	Behaviour of $\lambda_1(\mathbf{U}) = u - a$	$\int_{\mathbf{U}_{\frac{1}{2}}}^{\mathbf{U}_0} \mathbf{A}^-(\mathbf{U}) d\mathbf{U}$
1	$u_1 - a \geq 0, u_{\frac{1}{2}} - a \geq 0$	$\mathbf{0}$
2	$u_1 - a \leq 0, u_{\frac{1}{2}} - a \leq 0$	$\mathbf{F}_1 - \mathbf{F}_{\frac{1}{2}}$
3	$u_1 - a \leq 0, u_{\frac{1}{2}} - a \geq 0$	$\mathbf{F}_1 - \mathbf{F}_{S1}$
4	$u_1 - a \geq 0, u_{\frac{1}{2}} - a \leq 0$	$\mathbf{F}_{S1} - \mathbf{F}_{\frac{1}{2}}$

Table 12.5. Integration along path $I_1(\mathbf{U})$ connecting $\mathbf{U}_{\frac{1}{2}}$ to $\mathbf{U}_1$ (O-ordering), $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

	$u_0 + a \geq 0$ $u_1 - a \geq 0$	$u_0 + a \geq 0$ $u_1 - a \leq 0$	$u_0 + a \leq 0$ $u_1 - a \leq 0$	$u_0 + a \leq 0$ $u_1 - a \geq 0$
$u_{\frac{1}{2}} - a \geq 0$	$\mathbf{F}_0$	$\mathbf{F}_0 + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{S0} + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{S0}$
$u_{\frac{1}{2}} - a \leq 0$ $u_{\frac{1}{2}} + a \geq 0$	$\mathbf{F}_0 + \mathbf{F}_{S1}$ $-\mathbf{F}_{\frac{1}{2}}$	$\mathbf{F}_0 + \mathbf{F}_1$ $-\mathbf{F}_{\frac{1}{2}}$	$\mathbf{F}_{S0} + \mathbf{F}_1$ $-\mathbf{F}_{\frac{1}{2}}$	$\mathbf{F}_{S0} + \mathbf{F}_{S1}$ $-\mathbf{F}_{\frac{1}{2}}$
$u_{\frac{1}{2}} + a \leq 0$	$\mathbf{F}_0 + \mathbf{F}_{S1}$ $-\mathbf{F}_{S0}$	$\mathbf{F}_0 + \mathbf{F}_1$ $-\mathbf{F}_{S0}$	$\mathbf{F}_1$	$\mathbf{F}_{S1}$

Table 12.6. Osher's intercell flux for the isothermal equations using O-ordering of integration paths, $\mathbf{F}_k = \mathbf{F}(\mathbf{U}_k)$

12.3 Osher's Scheme for the Euler Equations

Here we develop in detail the Osher scheme, with both P and O orderings, for the time-dependent Euler equations. We first consider the one-dimensional case

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = \mathbf{0}, \quad (12.58)$$

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}, \quad \mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix}. \quad (12.59)$$

Details of the Euler equations are found in Sect. 1.1 of Chap. 1 and Chap. 3. We require an expression for the intercell flux $\mathbf{F}_{i+\frac{1}{2}}$ in the explicit conservative formula

$$\mathbf{U}_i^{n+1} = \mathbf{U}_i^n + \frac{\Delta t}{\Delta x} [\mathbf{F}_{i-\frac{1}{2}} - \mathbf{F}_{i+\frac{1}{2}}]. \quad (12.60)$$

Recall that the Jacobian matrix $\mathbf{A}(\mathbf{U})$, see Sect. 3.1.2 of Chap. 3, has eigenvalues

$$\lambda_1 = u - a, \quad \lambda_2 = u, \quad \lambda_3 = u + a \quad (12.61)$$

and right eigenvectors

$$\mathbf{K}^{(1)} = \begin{bmatrix} 1 \\ u - a \\ H - ua \end{bmatrix}, \quad \mathbf{K}^{(2)} = \begin{bmatrix} 1 \\ u \\ \frac{1}{2}u^2 \end{bmatrix}, \quad \mathbf{K}^{(3)} = \begin{bmatrix} 1 \\ u + a \\ H + ua \end{bmatrix}. \quad (12.62)$$

It is instructive to relate Osher's scheme to the solution of the Riemann problem in the classical sense, see Chap. 4 for details. Fig. 12.1 shows the

structure of the solution of the Riemann problem with data $\mathbf{U}_0, \mathbf{U}_1$ in the $x-t$ plane. Also shown there are the partial integration paths $I_1(\mathbf{U}), I_2(\mathbf{U}), I_3(\mathbf{U})$, the intersection points $\mathbf{U}_{\frac{1}{3}}, \mathbf{U}_{\frac{2}{3}}$ and the sonic points $\mathbf{U}_{S0}, \mathbf{U}_{S1}$ in phase space; the illustrated paths follow the *P-ordering*. There are essentially two steps in obtaining the Osher flux formulae. First the intersection points $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$ are obtained. We identify these points with the states $\mathbf{U}_{*L}$ and $\mathbf{U}_{*R}$ in the solution of the Riemann problem, see Chaps. 4 and 9. The second step consists of evaluating the integral in (12.21), for instance, along the integration paths $I_k(\mathbf{U})$ to obtain the intercell flux.

12.3.1 Osher's Flux with P-Ordering

The physical or *P-ordering* of integration paths for the Euler equations is illustrated in Fig. 12.1. States $\mathbf{U}_0$ and $\mathbf{U}_{\frac{1}{3}}$ are connected by the partial integration path $I_1(\mathbf{U})$, which is taken to be tangential to the right eigenvector $\mathbf{K}^{(1)}(\mathbf{U})$ in (12.62). Similarly, $I_2(\mathbf{U})$ connects $\mathbf{U}_{\frac{1}{3}}$ to $\mathbf{U}_{\frac{2}{3}}$ and $I_3(\mathbf{U})$ connects $\mathbf{U}_{\frac{2}{3}}$ to $\mathbf{U}_1$. In the *P-ordering* the intersection points $\mathbf{U}_{\frac{1}{3}}, \mathbf{U}_{\frac{2}{3}}$ and the sonic points $\mathbf{U}_{S0}, \mathbf{U}_{S1}$ are obtained by using the physically correct Generalised Riemann Invariants; see Sect. 3.1.3 of Chap. 3 for details.

Intersection Points and Sonic Points. Effectively, the intersection points $\mathbf{U}_{\frac{1}{3}}, \mathbf{U}_{\frac{2}{3}}$ can be taken to be the solution of the Riemann problem with data $\mathbf{U}_0, \mathbf{U}_1$ in the conventional sense. In the spirit of Osher's scheme we obtain $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$ utilising the Generalised Riemann Invariants

$$I_L \equiv u + \frac{2a}{\gamma - 1} = \text{constant} \quad (12.63)$$

and

$$I_R \equiv u - \frac{2a}{\gamma - 1} = \text{constant} \quad (12.64)$$

to relate $\mathbf{U}_0$ to $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$ to $\mathbf{U}_1$ respectively. Recall that if the left and right non-linear waves are rarefaction waves then relations (12.63)–(12.64) are exact. These waves can be either shock or rarefactions and thus the derived intersection points $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$ are, in general, approximations. The underlying assumption is that in the solution of the Riemann problem with data $\mathbf{U}_0, \mathbf{U}_1$, both non-linear waves are rarefaction waves. This corresponds to the Two-Rarefaction approximation TRRS presented in Sect. 9.4.1 of Chap. 9. Using (12.63) across the left wave gives

$$u_* + \frac{2a_{\frac{1}{3}}}{\gamma - 1} = u_0 + \frac{2a_0}{\gamma - 1} \quad (12.65)$$

and use of (12.64) across the right wave gives

$$u_* - \frac{2a_{\frac{2}{3}}}{\gamma - 1} = u_1 - \frac{2a_1}{\gamma - 1}. \quad (12.66)$$

Here u_* is the common particle velocity for $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$. Recall that, see Chaps. 3 and 4, the pressure p_* is also common, that is

$$u_{\frac{1}{3}} = u_{\frac{2}{3}} = u_* = \text{constant}, \quad p_{\frac{1}{3}} = p_{\frac{2}{3}} = p_* = \text{constant}. \quad (12.67)$$

In addition to (12.65)–(12.66) the isentropic law applied to the left and right waves gives

$$a_{\frac{1}{3}} = a_0(p_*/p_0)^z, \quad a_{\frac{2}{3}} = a_1(p_*/p_1)^z, \quad (12.68)$$

with

$$z = \frac{\gamma - 1}{2\gamma}. \quad (12.69)$$

From (12.65) and (12.68) we obtain

$$u_* = u_0 - \frac{2a_0}{\gamma - 1} \left[\left(\frac{p_*}{p_0} \right)^z - 1 \right]. \quad (12.70)$$

Similarly, use of (12.66) and (12.68) gives

$$u_* = u_1 + \frac{2a_1}{\gamma - 1} \left[\left(\frac{p_*}{p_1} \right)^z - 1 \right]. \quad (12.71)$$

Solving for p_* and u_* gives

$$p_* = \left[\frac{a_0 + a_1 - (u_1 - u_0)(\gamma - 1)/2}{a_0/p_0^z + a_1/p_1^z} \right]^{\frac{1}{z}}, \quad (12.72)$$

$$u_* = \frac{Hu_0/a_0 + u_1/a_1 + 2(H - 1)/(\gamma - 1)}{H/a_0 + 1/a_1}, \quad (12.73)$$

with

$$H = (p_0/p_1)^z.$$

The density values $\rho_{\frac{1}{3}}, \rho_{\frac{2}{3}}$ could be obtained from (12.68)–(12.69) or more directly as

$$\rho_{\frac{1}{3}} = \rho_0 \left(\frac{p_*}{p_0} \right)^{\frac{1}{\gamma}}, \quad \rho_{\frac{2}{3}} = \rho_1 \left(\frac{p_*}{p_1} \right)^{\frac{1}{\gamma}}. \quad (12.74)$$

The complete solution for $\mathbf{U}_{\frac{1}{3}}, \mathbf{U}_{\frac{2}{3}}$ is given by (12.72)–(12.74). This is identical to the two-rarefaction approximation of Sect. 9.4.1 of Chap. 9. The computation of the sonic points $\mathbf{U}_{S0}$ (left wave) and $\mathbf{U}_{S1}$ (right wave) is performed by first enforcing the sonic conditions $\lambda_1 = u - a = 0$ and $\lambda_3 = u + a = 0$, respectively and then applying the *corresponding* Generalised Riemann Invariants. The solution for the left sonic point is

$$\left. \begin{aligned} u_{S0} &= \frac{\gamma-1}{\gamma+1}u_0 + \frac{2a_0}{\gamma+1}, & a_{S0} &= u_{S0}, \\ \rho_{S0} &= \rho_0 \left(\frac{a_{S0}}{a_0} \right)^{\frac{2}{\gamma-1}}, & p_{S0} &= p_0 \left(\frac{\rho_{S0}}{\rho_0} \right)^{\gamma}. \end{aligned} \right\} \quad (12.75)$$

For the right sonic point the solution is

$$\left. \begin{aligned} u_{S1} &= \frac{\gamma-1}{\gamma+1} u_1 - \frac{2a_1}{\gamma+1}, & a_{S1} &= -u_{S1}, \\ \rho_{S1} &= \rho_1 \left(\frac{a_{S1}}{a_1} \right)^{\frac{2}{\gamma-1}}, & p_{S1} &= p_1 \left(\frac{\rho_{S1}}{\rho_1} \right)^{\gamma}. \end{aligned} \right\} \quad (12.76)$$

Integration Along Partial Paths. We adopt expression (12.21) for the Osher intercell flux

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}_0 + \int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U},$$

where the integral in phase space along the path

$$I(\mathbf{U}) = I_1(\mathbf{U}) \cup I_2(\mathbf{U}) \cup I_3(\mathbf{U})$$

gives

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}_0 + \int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{3}}} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} + \int_{\mathbf{U}_{\frac{1}{3}}}^{\mathbf{U}_{\frac{2}{3}}} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} + \int_{\mathbf{U}_{\frac{2}{3}}}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U}. \quad (12.77)$$

The integration along each partial path $I_k(\mathbf{U})$ is performed individually and the results are added to produce $\mathbf{F}_{i+\frac{1}{2}}$. The partial integrations are easily performed following the methodology presented in previous sections. For any given path there is only one wave involved. The left and right waves define genuinely non-linear fields, see Sect. 3.1.3 of Chap. 3, and the corresponding eigenvalue changes sign at most once, generating the sonic-point values. The second field (middle wave) is linearly degenerate, see Sect. 3.1.3 of Chap. 3, and thus the eigenvalue $\lambda_2(\mathbf{U})$ is constant along the path $I_2(\mathbf{U})$.

The integration results for each path are given in Table 12.7 and are labelled *A*, *B* and *C* respectively. To obtain the intercell flux $\mathbf{F}_{i+\frac{1}{2}}$ the integral terms in (12.77) must be selected according to the behaviour of the eigenvalue $\lambda_k(\mathbf{U})$ along the path $I_k(\mathbf{U})$. As seen in Table 12.7 there are 32 possible combinations i, j, l ; of these only 16 are realisable, which are tabulated as in Table 12.8. Here, in order to identify the correct expression for the resulting intercell flux $\mathbf{F}_{i+\frac{1}{2}}$, we split conditions on the intersection points, first column, from conditions on the data, top row.

12.3.2 Osher's Flux with O-Ordering

As originally presented, Osher's scheme uses the O-ordering of integration paths; this is precisely the opposite of the P-ordering described previously. The approach is used consistently to determine the intersection points, the sonic points and for performing the integration in (12.77). Fig. 12.4 illustrates the O-ordering of Osher's scheme as applied to a 3×3 system, such as the

$A :$	Behaviour of $\lambda_1(\mathbf{U}) = u - a$	$\int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{3}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U}$
1	$u_0 - a_0 \geq 0, \quad u_* - a_{\frac{1}{3}} \geq 0$	$\mathbf{0}$
2	$u_0 - a_0 \leq 0, \quad u_* - a_{\frac{1}{3}} \leq 0$	$\mathbf{F}_{\frac{1}{3}} - \mathbf{F}_0$
3	$u_0 - a_0 \geq 0, \quad u_* - a_{\frac{1}{3}} \leq 0$	$\mathbf{F}_{\frac{1}{3}} - \mathbf{F}_{S0}$
4	$u_0 - a_0 \leq 0, \quad u_* - a_{\frac{1}{3}} \geq 0$	$\mathbf{F}_{S0} - \mathbf{F}_0$
$B :$	Behaviour of $\lambda_2(\mathbf{U}) = u$	$\int_{\mathbf{U}_{\frac{1}{3}}}^{\mathbf{U}_{\frac{2}{3}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U}$
1	$u_* \geq 0$	$\mathbf{0}$
2	$u_* < 0$	$\mathbf{F}_{\frac{2}{3}} - \mathbf{F}_{\frac{1}{3}}$
$C :$	Behaviour of $\lambda_3(\mathbf{U}) = u + a$	$\int_{\mathbf{U}_{\frac{2}{3}}}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U}$
1	$u_1 + a_1 \geq 0, \quad u_* + a_{\frac{2}{3}} \geq 0$	$\mathbf{0}$
2	$u_1 + a_1 \leq 0, \quad u_* + a_{\frac{2}{3}} \leq 0$	$\mathbf{F}_1 - \mathbf{F}_{\frac{2}{3}}$
3	$u_* + a_{\frac{2}{3}} \geq 0, \quad u_1 + a_1 \leq 0$	$\mathbf{F}_1 - \mathbf{F}_{S1}$
4	$u_* + a_{\frac{2}{3}} \leq 0, \quad u_1 + a_1 \geq 0$	$\mathbf{F}_{S1} - \mathbf{F}_{\frac{2}{3}}$

Table 12.7. Integration along partial paths for the Euler equations following the P-ordering, $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

	$u_0 - a_0 \geq 0$ $u_1 + a_1 \geq 0$	$u_0 - a_0 \geq 0$ $u_1 + a_1 \leq 0$	$u_0 - a_0 \leq 0$ $u_1 + a_1 \geq 0$	$u_0 - a_0 \leq 0$ $u_1 + a_1 \leq 0$
$u^* \geq 0$ $u^* - a_{\frac{1}{3}} \geq 0$	$\mathbf{F}_0$	$\mathbf{F}_0 + \mathbf{F}_1$ $-\mathbf{F}_{S1}$	$\mathbf{F}_{S0}$	$\mathbf{F}_{S0} - \mathbf{F}_{S1}$ $+\mathbf{F}_1$
$u^* \geq 0$ $u^* - a_{\frac{1}{3}} \leq 0$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_{\frac{1}{3}}$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_{\frac{1}{3}} - \mathbf{F}_{S1}$ $+\mathbf{F}_1$	$\mathbf{F}_{\frac{1}{3}}$	$\mathbf{F}_1 + \mathbf{F}_{\frac{1}{3}}$ $-\mathbf{F}_{S1}$
$u^* \leq 0$ $u^* + a_{\frac{2}{3}} \geq 0$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_{\frac{2}{3}}$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_{\frac{2}{3}} - \mathbf{F}_{S1}$ $+\mathbf{F}_1$	$\mathbf{F}_{\frac{2}{3}}$	$\mathbf{F}_{\frac{2}{3}} - \mathbf{F}_{S1}$ $+\mathbf{F}_1$
$u^* \leq 0$ $u^* + a_{\frac{2}{3}} \leq 0$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_{S1}$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_1$	$\mathbf{F}_{S1}$	$\mathbf{F}_1$

Table 12.8. Osher's flux formulae for the Euler equations using P-ordering of integration paths, $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

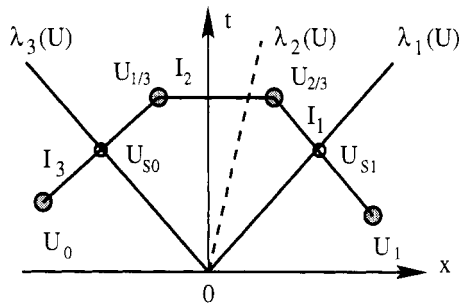


Fig. 12.4. Configuration of integration paths $I_k(\mathbf{U})$, intersection points $\mathbf{U}_{\frac{1}{3}}$, $\mathbf{U}_{\frac{2}{3}}$ and sonic points $\mathbf{U}_{S0}$, $\mathbf{U}_{S1}$ in physical space x - t for a 3×3 system, following the O-ordering

one-dimensional Euler equations. We combine the configuration for the integration paths, the intersection points and the sonic points in phase space with the Riemann problem solution with data $\mathbf{U}_0, \mathbf{U}_1$ in physical space $x-t$. The O-ordering can now be interpreted as assigning the eigenvalue $\lambda_3(\mathbf{U})$ and eigenvector $\mathbf{K}^{(3)}(\mathbf{U})$ to the wave family with eigenvalue $\lambda_1(\mathbf{U})$ and eigenvector $\mathbf{K}^{(1)}(\mathbf{U})$ (left wave), and vice-versa.

Intersection Points and Sonic Points. First, the intersection points $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$ are determined by using Generalised Riemann Invariants, as done in Sect. 12.1.3 with P-ordering. The difference is that $\mathbf{U}_0$ and $\mathbf{U}_{\frac{1}{3}}$ are connected using the *right* Riemann Invariant and $\mathbf{U}_{\frac{2}{3}}$ is connected to $\mathbf{U}_1$ using the *left* Riemann Invariant. Thus we write

$$I_R \equiv u_* - \frac{2}{\gamma-1} a_{\frac{1}{3}} = u_0 - \frac{2}{\gamma-1} a_0, \quad (12.78)$$

$$I_L \equiv u_* + \frac{2}{\gamma-1} a_{\frac{2}{3}} = u_1 + \frac{2}{\gamma-1} a_1. \quad (12.79)$$

See (12.63)–(12.66). Again, we assume u_* and p_* are the common particle velocity and pressure at points $\mathbf{U}_{\frac{1}{3}}$ and $\mathbf{U}_{\frac{2}{3}}$. Use of (12.68) into (12.78) gives

$$u_* = u_0 + \frac{2}{(\gamma-1)} a_0 \left[\left(\frac{p_*}{p_0} \right)^z - 1 \right], \quad (12.80)$$

and use of (12.68) into (12.79) produces

$$u_* = u_1 - \frac{2}{(\gamma-1)} a_1 \left[\left(\frac{p_*}{p_1} \right)^z - 1 \right]. \quad (12.81)$$

Solving for p_* gives

$$p_* = \left[\frac{a_0 + a_1 + (u_1 - u_0)(\gamma-1)/2}{a_0/p_0^z + a_1/p_1^z} \right]^{\frac{1}{z}}. \quad (12.82)$$

Compare with solution (12.72) using P-ordering. Equations (12.80) and (12.81) can be rearranged as

$$p_* = p_0 \left[\frac{\gamma-1}{2a_0} (u_* - u_0) + 1 \right]^{\frac{1}{z}}, \quad (12.83)$$

$$p_* = p_1 \left[\frac{\gamma-1}{2a_1} (u_1 - u_*) + 1 \right]^{\frac{1}{z}}, \quad (12.84)$$

whose solution for u_* is

$$u_* = \frac{H u_0 / a_0 + u_1 / a_1 - 2(H-1)/(\gamma-1)}{H/a_0 + 1/a_1}, \quad (12.85)$$

with

$$H = (p_0/p_1)^z, \quad z = \frac{\gamma-1}{2\gamma}.$$

Compare with solution (12.73) using the P-ordering. The solution for $\rho_{\frac{1}{3}}$ and $\rho_{\frac{2}{3}}$ is, using the isentropic law,

$$\rho_{\frac{1}{3}} = \rho_0(p_*/p_0)^{\frac{1}{\gamma}}, \quad \rho_{\frac{2}{3}} = \rho_1(p_*/p_1)^{\frac{1}{\gamma}}. \quad (12.86)$$

To find the sonic points $\mathbf{U}_{S0}$ and $\mathbf{U}_{S1}$ we first connect $\mathbf{U}_0$ to $\mathbf{U}_{\frac{1}{3}}$ via the *right* Riemann Invariant to obtain

$$u_{S0} = \frac{2}{\gamma-1}a_{S0} + u_0 - \frac{2}{\gamma-1}a_0.$$

Then by enforcing the sonic condition

$$\lambda_3(\mathbf{U}) = u_{S0} + a_{S0} = 0$$

along $I_3(\mathbf{U})$ and applying the isentropic law one obtains the solution

$$\left. \begin{aligned} u_{S0} &= \frac{\gamma-1}{\gamma+1}u_0 - \frac{2a_0}{\gamma+1}, & a_{S0} &= -u_{S0}, \\ \rho_{S0} &= \rho_0 \left(\frac{a_{S0}}{a_0} \right)^{\frac{2}{\gamma-1}}, & p_{S0} &= p_0 \left(\frac{\rho_{S0}}{\rho_0} \right)^{\gamma}. \end{aligned} \right\} \quad (12.87)$$

The solution for the right sonic point $\mathbf{U}_{S1}$ is

$$\left. \begin{aligned} u_{S1} &= \frac{\gamma-1}{\gamma+1}u_1 + \frac{2}{\gamma+1}a_1, & a_{S1} &= u_{S1}, \\ \rho_{S1} &= \rho_1 \left(\frac{a_{S1}}{a_1} \right)^{\frac{2}{\gamma-1}}, & p_{S1} &= p_1 \left(\frac{\rho_{S1}}{\rho_1} \right)^{\gamma}. \end{aligned} \right\} \quad (12.88)$$

Integration Along Partial Paths. To compute Osher's intercell flux

$$\mathbf{F}_{i+\frac{1}{2}} = \mathbf{F}_0 + \int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U}, \quad (12.89)$$

we need to find the integral

$$\int_{\mathbf{U}_0}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U} = \int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{3}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U} + \int_{\mathbf{U}_{\frac{1}{3}}}^{\mathbf{U}_{\frac{2}{3}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U} + \int_{\mathbf{U}_{\frac{2}{3}}}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U})d\mathbf{U}. \quad (12.90)$$

Using the O-ordering one has

$$\int_{\mathbf{U}_0}^{\mathbf{U}_{\frac{1}{3}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U} = \int_{I_3(\mathbf{U})} \mathbf{A}^-(\mathbf{U})d\mathbf{U}, \quad (12.91)$$

$$\int_{\mathbf{U}_{\frac{1}{3}}}^{\mathbf{U}_{\frac{2}{3}}} \mathbf{A}^-(\mathbf{U})d\mathbf{U} = \int_{I_2(\mathbf{U})} \mathbf{A}^-(\mathbf{U})d\mathbf{U}, \quad (12.92)$$

$$\int_{\mathbf{U}_{\frac{2}{3}}}^{\mathbf{U}_1} \mathbf{A}^-(\mathbf{U}) d\mathbf{U} = \int_{I_1(\mathbf{U})} \mathbf{A}^-(\mathbf{U}) d\mathbf{U}. \quad (12.93)$$

The evaluation of the three terms in (12.90) according to (12.91)–(12.93) is given in Table 12.9, by cases labelled A, B and C respectively. The first column contains the sub-case numbers. The second column contains the behaviour of the eigenvalue along the corresponding partial path. For instance in case A one considers the sign of the eigenvalue $\lambda_3(\mathbf{U})$ along the path $I_3(\mathbf{U})$ joining the points $\mathbf{U}_0$ and $\mathbf{U}_{\frac{2}{3}}$. The third column contains the resulting integral in (12.91)–(12.93). There are 32 combinations, of which only 16 are realisable.

Just to illustrate the method of analysis, consider first the combination (A_1, B_1, C_1) , that is

$$A_1 : u_0 + a_0 \geq 0, \quad u_* + a_{\frac{2}{3}} \geq 0,$$

$$B_1 : u_* \geq 0,$$

$$C_1 : u_1 - a_1 \geq 0, \quad u_* - a_{\frac{2}{3}} \geq 0.$$

As all eigenvalues are positive throughout, the integrals are all zero. The combination (A_1, B_2, C_1) gives

$$A_1 : u_0 + a_0 \geq 0, \quad u_* + a_{\frac{2}{3}} \geq 0,$$

$$B_2 : u_* \leq 0,$$

$$C_1 : u_1 - a_1 \geq 0, \quad u_* - a_{\frac{2}{3}} \geq 0.$$

Clearly these conditions cannot be satisfied simultaneously, $u_* \leq 0$ contradicts $u_* - a_{\frac{2}{3}} \geq 0$, as $a_{\frac{2}{3}} > 0$.

The Osher intercell flux with O-ordering is given in table 12.10. The logic involved can be organised so as to test conditions on the data points (top row) and conditions on the intersection points (first column). The Osher's flux formulae with P and O ordering given by Tables 12.8 and 12.10 respectively, are valid for any 3×3 non-linear hyperbolic system. The specific properties of a particular system enter in the determination of the sonic and intersection points only. The formulae can be directly applied, for instance, to the split two-dimensional shallow water equations.

12.3.3 Remarks on Path Orderings

It is useful to compare the pressure solutions (12.72) and (12.82) when computing the intersection points using the P and O orderings for the integration paths. Let us redefine the respective solutions as

$$p_*^{(P)} = \left[\frac{a_0 + a_1 - \Delta u(\gamma - 1)/2}{a_0/p_0^2 + a_1/p_1^2} \right]^{1/z} \quad (12.94)$$

$A :$	Behaviour of $\lambda_3(\mathbf{U}) = u - a$	$\int_{U_0}^{U_{\frac{1}{3}}} \mathbf{A}^-(\mathbf{U}) dU$
1	$u_0 + a_0 \geq 0, \quad u_* + a_{\frac{1}{3}} \geq 0$	$\mathbf{0}$
2	$u_0 + a_0 \leq 0, \quad u_* + a_{\frac{1}{3}} \leq 0$	$\mathbf{F}_{\frac{1}{3}} - \mathbf{F}_0$
3	$u_0 + a_0 \geq 0, \quad u_* + a_{\frac{1}{3}} \leq 0$	$\mathbf{F}_{\frac{1}{3}} - \mathbf{F}_{S0}$
4	$u_0 + a_0 \leq 0, \quad u_* + a_{\frac{1}{3}} \geq 0$	$\mathbf{F}_{S0} - \mathbf{F}_0$
$B :$	Behaviour of $\lambda_2(\mathbf{U}) = u$	
1	$u_* \geq 0$	$\mathbf{0}$
2	$u_* \leq 0$	$\mathbf{F}_{\frac{2}{3}} - \mathbf{F}_{\frac{1}{3}}$
$C :$	Behaviour of $\lambda_1(\mathbf{U}) = u - a$	
1	$u_1 - a_1 \geq 0, \quad u_* - a_{\frac{2}{3}} \geq 0$	$\mathbf{0}$
2	$u_1 - a_1 \leq 0, \quad u_* - a_{\frac{2}{3}} \leq 0$	$\mathbf{F}_1 - \mathbf{F}_{\frac{2}{3}}$
3	$u_1 - a_1 \geq 0, \quad u_* - a_{\frac{2}{3}} \leq 0$	$\mathbf{F}_{S1} - \mathbf{F}_{\frac{2}{3}}$
4	$u_1 - a_1 \leq 0, \quad u_* - a_{\frac{2}{3}} \geq 0$	$\mathbf{F}_1 - \mathbf{F}_{S1}$

Table 12.9. Osher's flux for the Euler equations. Evaluation of integral following O-ordering, $\mathbf{F}_k \equiv \mathbf{F}(\mathbf{U}_k)$

	$u_0 + a_0 \geq 0$ $u_1 - a_1 \geq 0$	$u_0 + a_0 \geq 0$ $u_1 - a_1 \leq 0$	$u_0 + a_0 \leq 0$ $u_1 - a_1 \geq 0$	$u_0 + a_0 \leq 0$ $u_1 - a_1 \leq 0$
$u_* + a_{\frac{1}{3}} \leq 0$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_{S1}$	$\mathbf{F}_0 - \mathbf{F}_{S0}$ $+\mathbf{F}_1$	$\mathbf{F}_{S1}$	$\mathbf{F}_1$
$u_* + a_{\frac{1}{3}} \geq 0$ $u_* \leq 0$	$\mathbf{F}_0 - \mathbf{F}_{\frac{1}{3}}$ $+\mathbf{F}_{S1}$	$\mathbf{F}_0 - \mathbf{F}_{\frac{1}{3}}$ $+\mathbf{F}_1$	$\mathbf{F}_{S0} - \mathbf{F}_{\frac{1}{3}}$ $+\mathbf{F}_{S1}$	$\mathbf{F}_{S0} - \mathbf{F}_{\frac{1}{3}}$ $+\mathbf{F}_1$
$u_* - a_{\frac{2}{3}} \geq 0$	$\mathbf{F}_0$	$\mathbf{F}_0 - \mathbf{F}_{S1}$ $+\mathbf{F}_1$	$\mathbf{F}_{S0}$	$\mathbf{F}_{S0} - \mathbf{F}_{S1}$ $+\mathbf{F}_1$
$u_* - a_{\frac{2}{3}} \leq 0$ $u_* \geq 0$	$\mathbf{F}_0 - \mathbf{F}_{\frac{2}{3}}$ $+\mathbf{F}_{S1}$	$\mathbf{F}_0 - \mathbf{F}_{\frac{2}{3}}$ $+\mathbf{F}_1$	$\mathbf{F}_{S0} - \mathbf{F}_{\frac{2}{3}}$ $+\mathbf{F}_{S1}$	$\mathbf{F}_{S0} - \mathbf{F}_{\frac{2}{3}}$ $+\mathbf{F}_1$

Table 12.10. Osher's flux formulae for the Euler equations using O-ordering of integration paths, $\mathbf{F}_k = \mathbf{F}(\mathbf{U}_k)$

and

$$p_*^{(O)} = \left[\frac{a_0 + a_1 + \Delta u(\gamma - 1)/2}{a_0/p_0^z + a_1/p_1^z} \right]^{1/z}, \quad (12.95)$$

where $\Delta u = u_1 - u_0$ is the velocity difference in the data. The reader is encouraged to review Sect. 4.3.1 of Chap. 4, in which a detailed discussion is given on the influence of Δu on the solution for pressure in the exact solution. First we note that $p_*^{(P)} = p_*^{(O)}$ when $\Delta u = 0$ and that for $\Delta u \neq 0$ the two solutions are not only different but more importantly, they have *very different behaviour*. This does not seem to matter too much in practical computations, except for special but important situations. There are at least two cases that deserve special attention.

The case $\Delta u \gg 0$, see Sect. 4.3.1 of Chap. 4, is associated with strong rarefaction waves. In fact there is a limit for which the pressure p_* becomes negative and is associate with the *pressure positivity condition* stated in Sect. 4.3.1 of Chap. 4. In the *incipient cavitation case* the pressure is 0. The solution $p_*^{(P)}$ will correctly reflect this physical situation of low-density flow, including the detection of *vacuum*. The author is not aware of this having been exploited in the context of Osher's Riemann solver with P-ordering. On the other hand, the solution $p_*^{(O)}$ for $\Delta u \gg 0$ obtained with the O-ordering, will give unrealistically large values for the pressure at the intersection points, which is more consistent with the presence of strong shock waves, rather than strong

rarefaction waves. Such large pressure values will lead the Osher scheme to be very inaccurate or simply to fail for low-density flows, just as linearised Riemann solvers do; see Einfeldt et. al. [118] for a discussion on numerical difficulties for low-density flows. We illustrate this point through Test 2 of Table 12.11, for which the scheme actually fails.

The case $\Delta u < 0$, see Sect. 4.3.1 of Chap. 4, is associated with strong shock waves. Again the P-ordering solution (12.94) will correctly reflect this. However the O-ordering solution (12.95) will not. More importantly, there will be a limiting strong shock situation for which the O-ordering pressure is *undefined* and the scheme will again fail; see Test 5 in Table 12.11. The failure condition in this case is analytic, namely

$$a_0 + a_1 + \Delta u(\gamma - 1)/2 \leq 0. \quad (12.96)$$

It is paradoxical that the scheme fails in the presence of strong shocks, which is consistent with large pressure, through a pressure solution $p_*^{(O)}$ that is undefined for being so close to vacuum conditions. One could devise some sort of fix to remedy this situation. One possibility is to abandon the O-ordering altogether or switch to the P-ordering locally, in a kind of adaptive ordering.

12.3.4 The Split Three-Dimensional Case

The extension of the Osher scheme to two and three-dimensional problems is straightforward. All methods considered here require expressions for the split fluxes. In the x -split, three dimensional case, for instance, we require in addition the y and z momentum flux components ρuv and ρuw . In turn, this requires the extra components ρv and ρw in the vector of conserved variables $\mathbf{U}$. These new components will be needed in the intersection points and sonic points. But, as seen in Sect. 4.8 of Chap. 4, the exact solution for v and w in the Riemann problem is given by

$$q(x, t) = \begin{cases} q_L & \text{if } x/t < u_* \\ q_R & \text{if } x/t > u_* \end{cases}, \quad (12.97)$$

where $q = v$ and $q = w$. The approximate solution for the tangential components of velocity obtained from using Generalised Riemann Invariants in the Osher scheme preserves the form of the exact solution (12.97), the only approximation being that of the normal velocity component u_* . Therefore, for computing $\mathbf{U}_{S0}$ and $\mathbf{U}_{\frac{1}{3}}$ we take v_L and w_L ; for computing $\mathbf{U}_{S1}$ and $\mathbf{U}_{\frac{2}{3}}$ we take v_R and w_R . Solution (12.97) ensures that contact waves and shear waves (and shear layers in Navier–Stokes applications) are well resolved by Osher's scheme, a property that is common to the exact Riemann solver of Chap. 6, the approximate-state Riemann solvers of Chap. 9, the HLLC Riemann solver of Chap. 10 and Roe's solver of Chap. 11.

12.4 Numerical Results and Discussion

Here we illustrate the performance of Godunov's first-order method used in conjunction with Osher-type approximate Riemann solvers. We select six test problems for the one-dimensional time dependent Euler equations for ideal gases with $\gamma = 1.4$; these have exact solutions. In all chosen tests, data consists of two constant states $\mathbf{W}_L = [\rho_L, u_L, p_L]^T$ and $\mathbf{W}_R = [\rho_R, u_R, p_R]^T$, separated by a discontinuity at a position $x = x_0$. The states $\mathbf{W}_L$ and $\mathbf{W}_R$ are given in Table 12.11. The exact and numerical solutions are found in the spatial domain $0 \leq x \leq 1$. The numerical solution is computed with $M = 100$ cells and the CFL condition is as for all previous computations, see Chap. 6; the chosen Courant number coefficient is $C_{\text{eff}} = 0.9$; boundary conditions are transmissive. The exact solutions were found by running the code *HE-EIRPEXACT* of the library *NUMERICA* [369] and the numerical solutions were obtained by running the code *HE-E1GODOSHER* of *NUMERICA*.

Test	ρ_L	u_L	p_L	ρ_R	u_R	p_R
1	1.0	0.75	1.0	0.125	0.0	0.1
2	1.0	-2.0	0.4	1.0	2.0	0.4
3	1.0	0.0	1000.0	1.0	0.0	0.01
4	5.99924	19.5975	460.894	5.99242	-6.19633	46.0950
5	1.0	-19.59745	1000.0	1.0	-19.59745	0.01
6	1.0	2.0	0.1	1.0	-2.0	0.1

Table 12.11. Data for six test problems with exact solution, for the time-dependent one dimensional Euler equations

Test 1 is a *modified* version of Sod's problem [318]; the solution has a right shock wave, a right travelling contact wave and a left *sonic* rarefaction wave; this test is useful in assessing the *entropy satisfaction* property of numerical methods. The solution of Test 2 consists of two symmetric rarefaction waves and a trivial contact wave; the *Star Region* between the non-linear waves is close to vacuum, which makes this problem a suitable test for assessing the performance of numerical methods for low-density flows. Test 3 is designed to assess the robustness and accuracy of numerical methods; its solution consists of a strong right shock wave of shock Mach number 198, a contact surface and a left rarefaction wave. Test 4 is also a very severe test, its solution consists of three strong discontinuities travelling to the right. A detailed discussion on the exact solution of the test problems is found in Chap. 4. Test 5 is also designed to test the robustness of numerical methods but the main reason for devising this test is to assess the ability of numerical methods to resolve *slowly-moving contact discontinuities*. The exact solution of Test 5 consists of a left rarefaction wave, a right-travelling shock wave and a *stationary* contact discontinuity. Test 6 simulates the collision of two uniform streams; the exact solution consists of two strong symmetric shock waves and a trivial

contact discontinuity. The purpose of Test 6 is to illustrate the fact that Osher's scheme with O-ordering will fail for a range of problems of this kind. For each test problem we select a convenient position x_0 of the initial discontinuity and the output time. These are stated in the legend of each figure displaying computational results.

The computed results for Tests 1, 3, 4 and 5 using the Godunov first-order method in conjunction with the Osher approximate Riemann solver with O-ordering are shown in Figs. 12.5–12.8. The scheme failed for Tests 2 and 6. The result of Fig. 12.5 is virtually identical to that of the Godunov method with an exact Riemann solver, Fig. 6.8 of Chap. 6; note also that both schemes give comparable performance in the vicinity of the sonic point. The results for Tests 3 and 4, shown in Figs. 12.6 and 12.7, obtained with O-ordering compare well with those of the exact Riemann solver, except for the fact that the slowly moving shock in Fig. 12.7 does not have the spurious oscillations that other Riemann solvers produce [280], [40], [13], [193]; this is a distinguishing property of the Osher approach.

Results for Tests 1 to 6 from the Osher scheme with P-ordering are shown in Figs. 12.9–12.14. The results for Tests 1 to 4 are, overall, as accurate as those obtained from an exact Riemann solver; the P-ordering scheme actually works for Tests 2 and 6, whereas O-ordering scheme does not; note however the spurious oscillations in Test 6 near the shocks, a feature of P-ordering schemes already noted by Osher. For Test 5 the P-ordering scheme gives the incorrect solution; compare Fig. 12.13 with Fig. 12.8 and note the scales. First, there are very large unphysical overshoots in density, velocity and pressure. More intriguing is the fact that the expected right-travelling shock does not propagate at all; no signal propagates to the right of the initial discontinuity at $x = 0.8$. As a matter of fact, it looks as if the shock wave does not actually form and the state behind the contact discontinuity has the incorrect values, see pressure and velocity plots. My numerical results have been confirmed by Dr N. Qin (private communication), who used an independently written code to solve Test 5.

Earlier we made some remarks concerning a modified Osher scheme based on an adaptive ordering. In view of the numerical results it seems as if O-ordering could be used in all cases except for the two situations in which the solution of the Riemann problem contains either two shocks or two rarefactions. Such situations could be identified reliably and cheaply by using any of the approximate state Riemann solvers of Chap. 9.

12.5 Extensions

The approximate Riemann solvers of this chapter may be applied in conjunction with the Godunov first-order upwind method presented in Chap. 6. Second-order Total Variation Diminishing (TVD) extensions of the schemes are presented in Chap. 13 for scalar problems and in Chap. 14 for non-linear

one dimensional systems. In chap. 15 we present techniques that allow the extension of these schemes to solve problems with source terms. In Chap. 16 we study techniques to extend the methods of this chapter to three-dimensional problems.

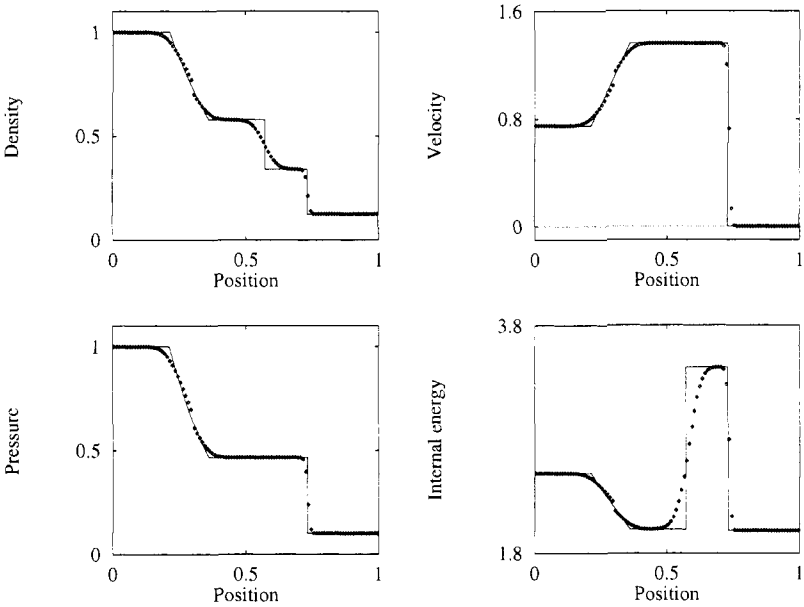


Fig. 12.5. Osher scheme with O-ordering applied to Test 1, with $x_0 = 0.3$. Numerical (symbol) and exact (line) solutions are compared at time 0.2 units

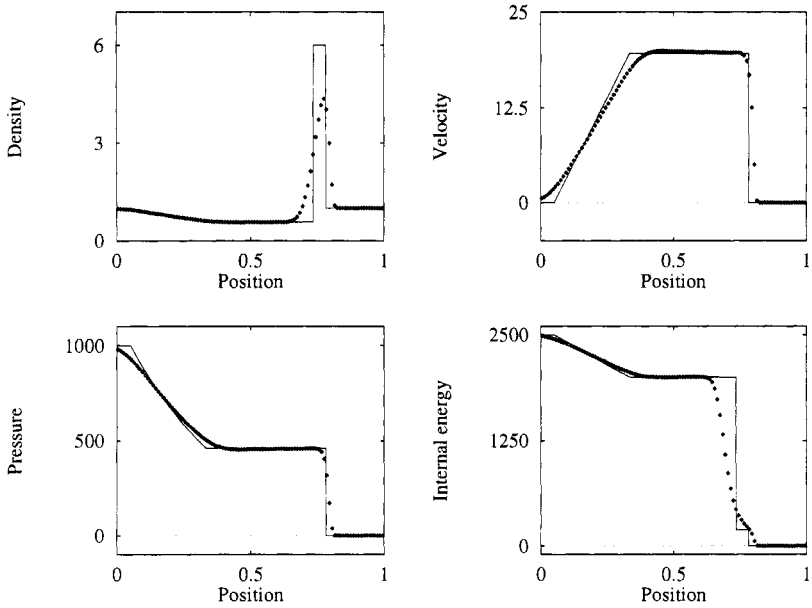


Fig. 12.6. Osher scheme with O-ordering applied to Test 3, with $x_0 = 0.5$. Numerical (symbol) and exact (line) solutions are compared at time 0.012 units

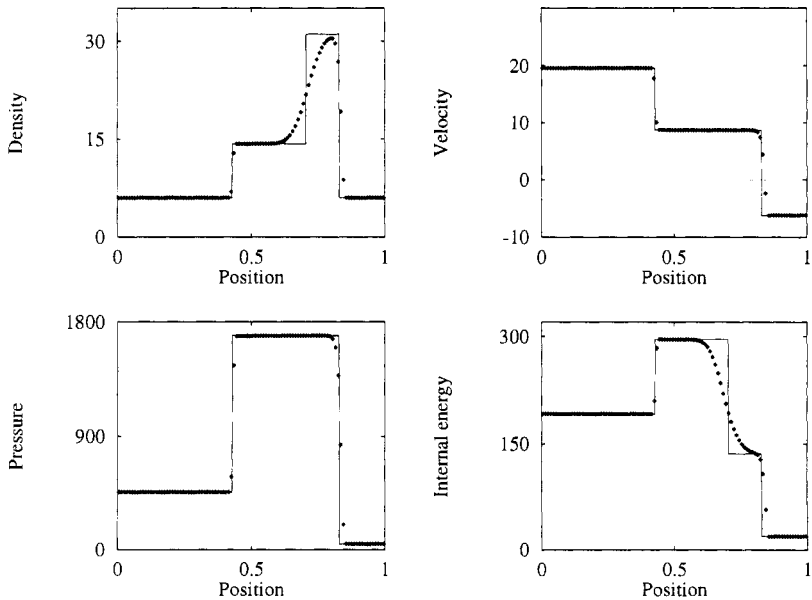


Fig. 12.7. Osher scheme with O-ordering applied to Test 4, with $x_0 = 0.4$. Numerical (symbol) and exact (line) solutions are compared at time 0.035 units

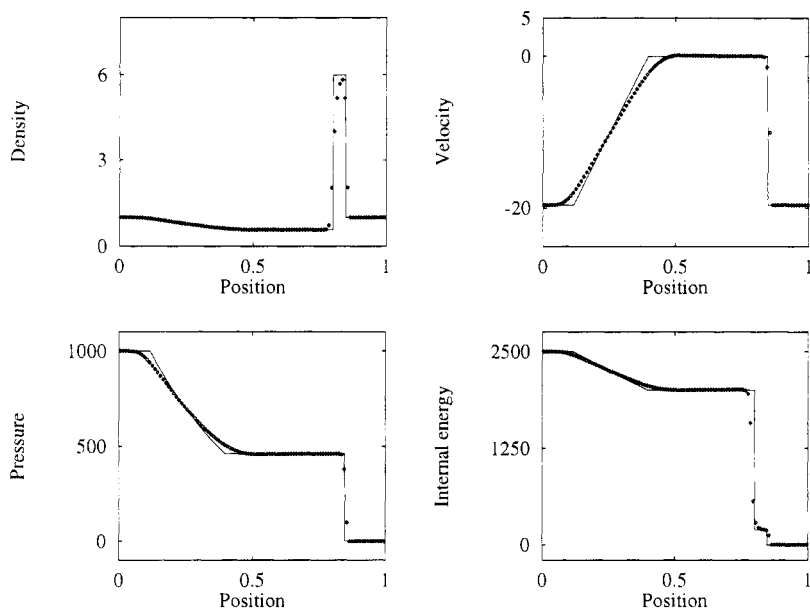


Fig. 12.8. Osher scheme with O-ordering applied to Test 5, with $x_0 = 0.8$. Numerical (symbol) and exact (line) solutions are compared at time 0.012 units

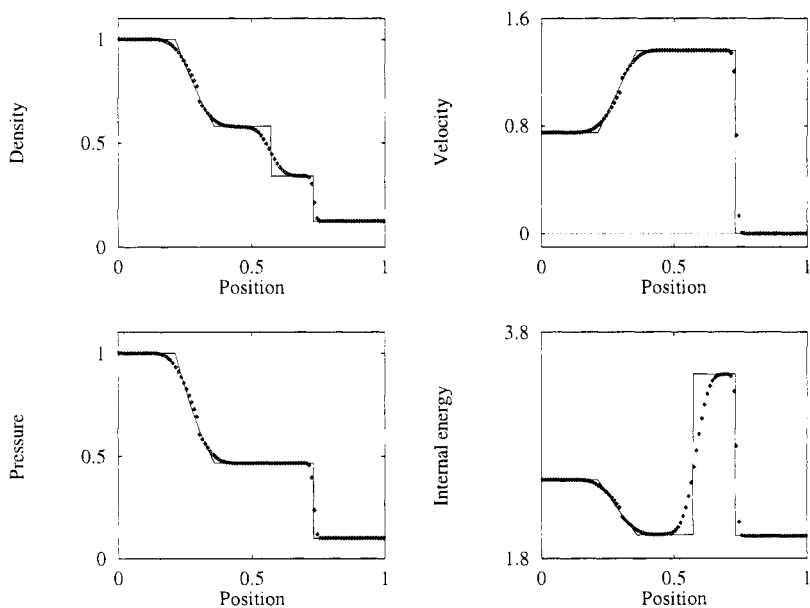


Fig. 12.9. Osher scheme with P-ordering applied to Test 1, with $x_0 = 0.3$. Numerical (symbol) and exact (line) solutions are compared at time 0.2 units

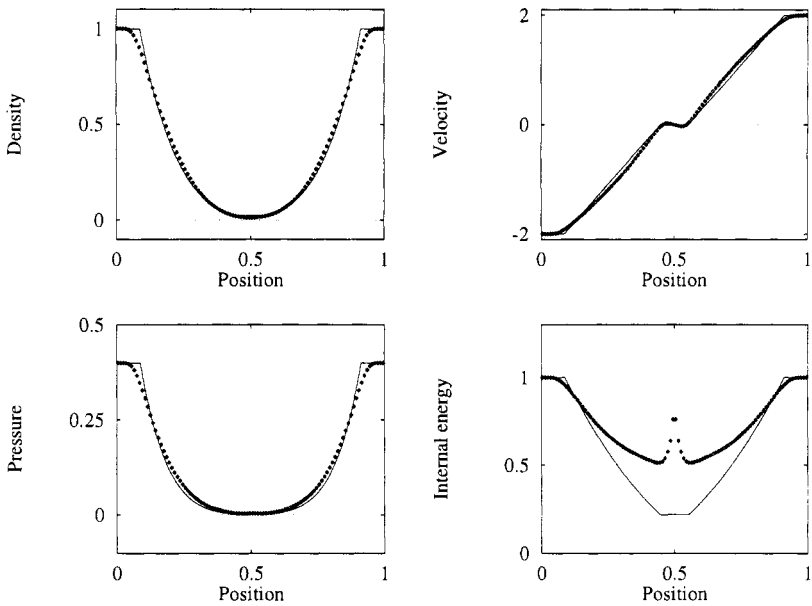


Fig. 12.10. Osher scheme with P-ordering applied to Test 2, with $x_0 = 0.3$. Numerical (symbol) and exact (line) solutions are compared at time 0.2 units

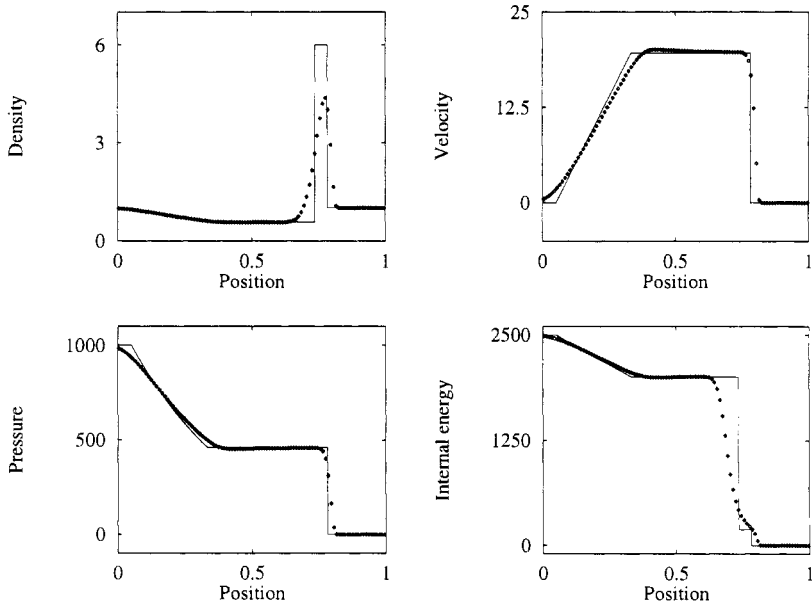


Fig. 12.11. Osher scheme with P-ordering applied to Test 3, with $x_0 = 0.5$. Numerical (symbol) and exact (line) solutions are compared at time 0.012 units

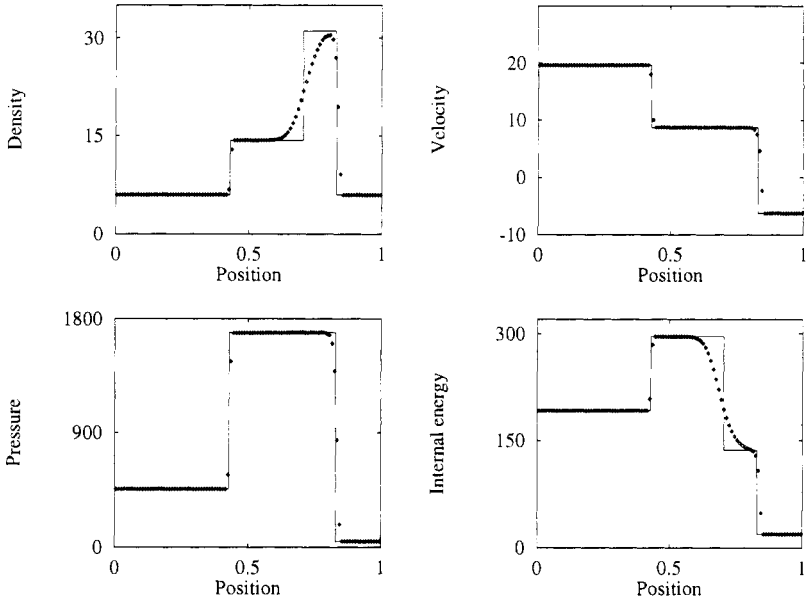


Fig. 12.12. Osher scheme with P-ordering applied to Test 4, with $x_0 = 0.4$. Numerical (symbol) and exact (line) solutions are compared at time 0.035 units

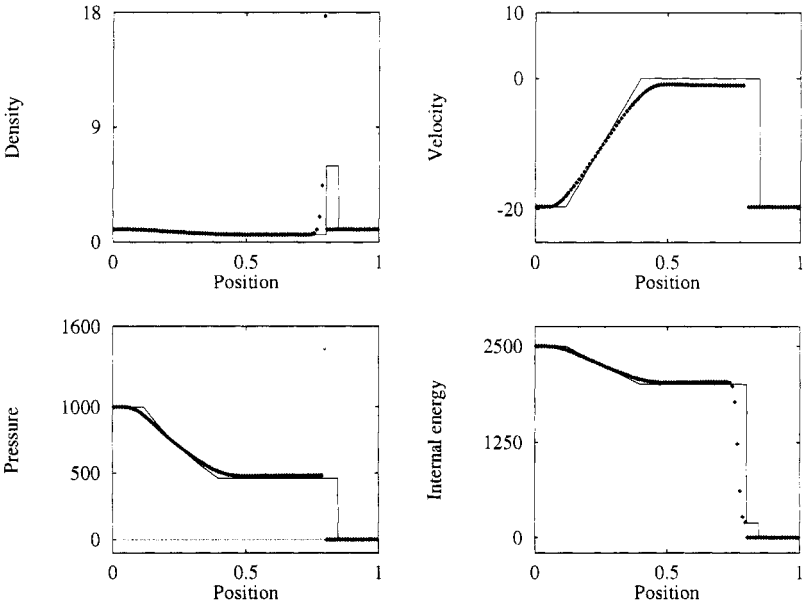


Fig. 12.13. Osher scheme with P-ordering applied to Test 5, with $x_0 = 0.8$. Numerical (symbol) and exact (line) solutions are compared at time 0.012 units

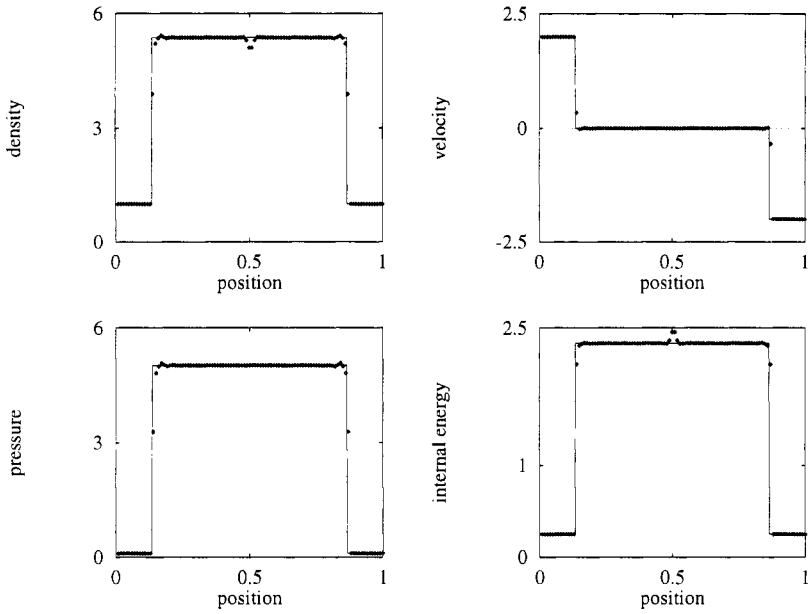


Fig. 12.14. Osher scheme with P-ordering applied to Test 6, with $x_0 = 0.5$. Numerical (symbol) and exact (line) solutions are compared at time 0.8 units